Molecular Phylogenetics and Evolution 148 (2020) 106811

Contents lists available at ScienceDirect

Molecular Phylogenetics and Evolution

journal homepage: www.elsevier.com/locate/ympev

Cryptic diversity and ancient diversification in the northern Atlantic Forest *Pristimantis* (Amphibia, Anura, Craugastoridae)

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ARTICLE INFO

Keywords:
Terrarana
Netropical region
Eocene
Anura

ABSTRACT

The Neotropical region is known both for its megadiverse fauna and for the significant deficiency of our knowledge on species limits. The Amazon and Atlantic Forest are the two most diverse and large rainforests in South America, and they harbor many groups of sister species and sister genera. The frog genus *Pristimantis* is the most speciose genus of terrestrial vertebrates with 546 species, but only three of them occur in the Atlantic forest. Herein, we investigate the diversification history and phylogenetic relationship among the Atlantic Forest *Pristimantis* lineages in a spatial-temporal framework, using mitochondrial and nuclear genes. Our results reveal a remarkable hidden diversity, with nine highly structure lineages that may correspond to undescribed species, with many cases of sympatry among these divergent lineages. Atlantic forest *Pristimantis* form a monophyletic group that started to diversify over 40 million years ago. This ancient group shows diversification events that remount the early bursts of the Atlantic Forest diversification history, as well as lineage diversification likely resulting from recent Pleistocene climatic fluctuations. Future work must concentrate in comparing these lineages under an integrative framework including morphology, advertisement calls and other ecology traits to confidently delimit species of *Pristimantis* in the Atlantic Forest.

1. Introduction

The Neotropical region is known worldwide for its astonishing levels of biodiversity and endemism (Myers et al., 2000), which are associated to a complex evolutionary history with many processes involved in the accumulation of species (Whitmore and Prance, 1987; Moritz et al., 2000; Antonelli and Sanmartín, 2011). Nevertheless, describing the Neotropical biota and its systematic relationships is not a trivial task, as many species groups have shortfalls on knowledge regarding species limits (Linnean shortfall) and evolutionary relationships (Darwinian shortfall) (Diniz-Filho et al., 2013; Hortal et al., 2015). However, advances on molecular systematics and the increasing of availability of molecular data have largely contributed to settle the biodiversity shortfalls, especially in hyperdiverse regions as the Atlantic Forest.

The highly diverse and threatened Atlantic Forest is on the world's top five biodiversity hotspots (Myers et al., 2000). The evolutionary history of this ecoregion has been investigated in many studies in the last years (Carnaval et al., 2009, 2014; Leite et al., 2016). Most advocated the role of Pleistocene climatic changes in driving the isolation of species and demographic fluctuations of populations (e.g. Cabanne et al., 2008; Carnaval et al., 2009; d'Horta et al., 2011; Batalha-Filho

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https://doi.org/10.1016/j.ympev.2020.106811
Received 1 October 2019; Received in revised form 12 March 2020; Accepted 18 March 2020
Available online 04 April 2020
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et al., 2012; Maldonado-Coelho, 2012; Batalha-Filho and Miyaki, 2016). Alternative hypotheses of tectonism (Batalha-Filho et al., 2010; Brunes et al., 2010; Thomé et al., 2010; Amaro et al., 2012) and population expansions into the exposed continental shelf during the Pleistocene (Leite et al., 2016) have also been evoked to explain species diversification in the Atlantic Forest. However, most studies on diversification of the Atlantic Forest species focused so far on central and southern species, with few studies of northern species (Silveira et al., 2019). Interestingly, recent ecological niche modeling revealed that the northern Atlantic Forest expanded its range during the last glacial maximum, in contrast to a contraction on its southern portion (Silveira et al., 2019). Therefore, studies focusing on the northern portion of the biome are still scarce and might reveal a diversification history distinct from the southern region.

Herein, we studied the population structure and diversification of Terrarana frogs endemic to the northern Atlantic Forest. The Terraranae Hedges, Duellman, and Heinicke, 2008 (emended from Terrarana by Duellman et al., 2016), an informal and unranked name for the taxon Brachycephaloidea Günther, 1858 (Padial et al., 2014), is one of several anurans' taxa widely distributed throughout the megadiverse Neotropical realm. The terraranans are New World direct-developing frogs and currently gather over 1150 species distributed from southern United States to southern South America (Frost, 2019). They currently comprise three families and two genera ranked as *incertae sedis* (Padial et al., 2014; Frost, 2019): Brachycephalidae Günther, 1858, Craugastoridae Hedges et al., 2008, Eleutherodactylidae Lutz, 1954, *Atopophrynus* Lynch and Ruiz-Carranza, 1982 and *Geobatrachus* Ruthven, 1915. The Craugastoridae genus *Pristimantis* Jiménez de la Espada, 1870 (Ceuthomantinae Heinicke et al., 2009) is the focus of the current study.

The genus *Pristimantis* is the most speciose genus among vertebrates (Waddell et al., 2018) and currently comprises 546 species distributed from Central America to southern South America (Frost, 2019). Throughout this wide-range geographical distribution, *Pristimantis* species can be found inhabiting different biomes and environments, but mainly rainforest. Only three species are currently known to inhabit the Atlantic Forest, all endemic to its northernmost portion (northeastern Brazil): *P. ramagii* (Boulenger, 1888), *P. paulodutrai* (Bokermann, 1975) and *P. vinhai* (Bokermann, 1975). *Pristimantis ramagii* occurs mainly north of the São Francisco River, and the other two species are restricted to the Atlantic Forest of Bahia state (but see Santana et al., 2016; Santos et al., 2016 for *P. vinhai* records for Paraíba and Pernambuco states). Remarkably, these Atlantic Forest *Pristimantis* comprise a monophyletic group in the *Pristimantis conspicillatus* species group (Canedo and Haddad, 2012), though the relationships inside the Atlantic Forest clade remain unresolved. The three species are considered as least concern according to IUCN criteria (Juncá and Carnaval, 2004; Pimenta and Juncá, 2004; Peixoto and Pimenta, 2004).

Frogs of the genus *Pristimantis* in northeastern Brazil are usual inhabitants of forests along the Atlantic coast (e.g., Juncá, 2006; Bastazini et al., 2007; Dias et al., 2014; Gondim-Silva et al., 2016; Oitaven et al., 2017; Mira-Mendes et al., 2018), but are also found in ciliary, altitudinal and dry forests within the Caatinga biome (e.g., Loebmann and Haddad, 2010; Xavier and Napoli, 2011; Garda et al., 2013). These species exhibit high intraspecific variation on color patterns, but similar external morphologies, which led to misidentifications in herpetological collections and publications (Napoli et al., 2009). Furthermore, those authors highlighted four additional issues regarding the incorrect specimens' identification: (1) succinct original description of *P. ramagii*; (2) insufficient diagnosis of *P. paulodutrai*; (3) morphological polymorphism within local populations; and (4) poor knowledge of intra- and interpopulation variation of *P. ramagii* and *P. paulodutrai*. Indeed, these difficulties and possible misidentifications were already pointed out by Carnaval and Bates (2007), which observed high genetic variation in population samples of *Pristimantis* gr. *ramagii* from enclaves of altitudinal forest patches among Caatinga, known as "Brejos de Altitude" (see Vanzolini, 1981 for definition of "Brejos de Altitude"), suggesting that it might be, in fact, a species complex. Finally, advertisement call parameters, which could elucidate the diversity and diagnosis of *Pristimantis* species from the Atlantic Forest, are incomplete in the literature (absence of *P. vinhai* advertisement call description) and lack information on geographical variation (*P. paulodutrai*, Bokermann, 1975; *P. ramagii*, Oitaven et al., 2017). Therefore, calls may be associated to wrong focal species (*P. ramagii*, Heyer and Carvalho, 2000), and species-level recognition is problematic due to our current poor knowledge on the complex call variation resulting from social interactions (M.F. Napoli personal observation).

In summary, similar color patterns of external body surfaces, barely distinguishable external morphologies, and incomplete knowledge on vocal repertoire have been hindering the identification of *Pristimantis* species of the Atlantic Forest, which could explain the low number of nominal species recognized up to now. This assumption has vast support in recent literature regarding anurans' species diversity, in which widely distributed cryptic nominal species were split into several candidate species often with restricted to very small geographical distribution ranges (e.g., Padial and De La Riva, 2009; Fouquet et al., 2012; Gehara et al., 2014; Rakotoarison et al., 2017; Jansen et al., 2019).

In this study, we investigate the diversification history and phylogenetic relationship among the Atlantic Forest *Pristimantis* lineages in a spatial-temporal framework, using mitochondrial and nuclear genes. We specifically addressed the following questions: (i) How many lineages exist along the geographical distribution of Atlantic Forest *Pristimantis*? (ii) Are there lineages that could be unequivocally associated to each available nominal *Pristimantis* species from the Atlantic Forest? (iii) Are there any lineages that could be considered as candidate species? (iv) When did the divergence among lineages take place?

2. Material and methods

2.1. Taxa sampling and molecular methods

We obtained 173 samples, from 57 localities (Table S1, in supplementary material; Fig. 1) encompassing the known range of all species of the Atlantic Forest *Pristimantis* clade, the type localities of the three recognized nominal species (*P. paulodutrai*, *P. ramagii* and *P. vinhai*), as well as isolated populations in forest enclaves at the Chapada Diamantina and northern Ceará state. Specimens from type localities were associated to nominal species only after examination of external morphology, external color patterns and advertisement calls, using for comparisons literature data (Boulenger, 1888; Bokermann, 1975; Napoli et al., 2009; Fonseca, 2011; Nascimento, 2018), advertisement calls (sound files housed at the Laboratory of Taxonomy and Natural History of Amphibians, Federal University of Bahia, Brazil) and preserved specimens deposited in the Natural History Museum of the Federal University of Bahia (Supplementary data 1, Supplementary material).

Total DNA was extracted from tissue samples (liver, muscle or digit) using the PureLink Genomic DNA extraction kit (Invitrogen). We obtained sequences for the mitochondrial gene NADH dehydrogenase subunit 2 (*ND2*) using primers ND2ALAR and ND2ELEUF1 (Carnaval and Bates, 2007) and three nuclear introns: Beta-crystallin (*CRYBA*) with primers Cryb1Ls and Cryb2Ls (Dolman and Phillips, 2004), 28S ribosomal RNA (*28S*) with primers 28SV and 28SSJ (Hillis and Dixon, 1991), and *X. laevis* MGC82783 protein Intron2 (*MGC82783*) with primers MVZ39 and MVZ40 (Bell et al., 2012). PCR for the mitochondrial gene consisted of 23 μL of PCR Supermix (Invitrogen) and 2 μL of template DNA. PCR mix (25 μL) for nuclear loci contained 2 μL of template DNA, 1X of Platinum Taq polymerase buffer (Invitrogen), 0.32 μM of dNTPs, 0.5 μM of each primer and 0.5 U of Platinum Taq polymerase (Invitrogen). PCR conditions were as follows: an initial denaturation step at 95 °C for 3 min, followed by 35 cycles at 95 °C for

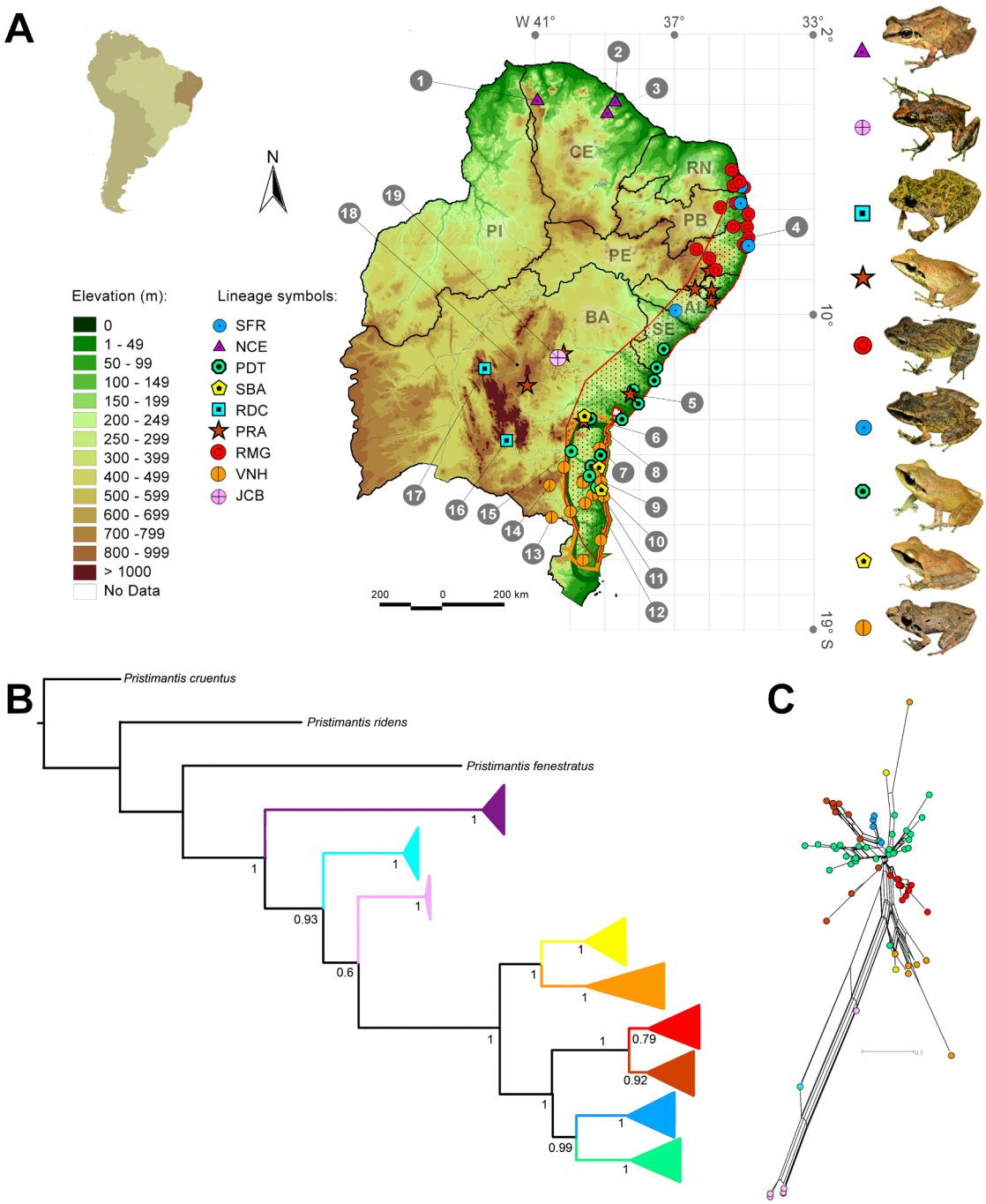

Fig. 1. Sampling localities, phylogenetic inference and nuclear multilocus network for *Pristimantis* from the Atlantic Forest (AF). Colors depict distinct clades in all figures. (A) Topographical map highlighting point locality sampling of each clade (lineage symbols), with photographs of a representative individual of each clade (photos by R. O Abreu and L. Menezes). Brazilian states: AL, Alagoas; BA, Bahia; CE, Ceará; PB, Paraíba; PE, Pernambuco; PI, Piauí; RN, Rio Grande do Norte; SE, Sergipe. Sampling localities (municipalities) referred in text: 1, Ubajara; 2, Maranguape; 3, Pacoti; 4, Igarassu (type locality of *P. ramagii*); 5, Catu; 6, Varzedo; 7, Igrapiúna; 8, Elísio Medrado; 9, Ilhéus (type locality of *P. paulodutrai* and *P. vinhai*); 10, Uruçuca; 11, Una; 12, Arataca; 13, Macarani; 14, Barra do Choça; 15, Jequié; 16, Rio de Contas; 17, Gentio do Ouro; 18, Chapada Diamantina Ecoregion; 19, Miguel Calmon. Shaded areas within polygons are geographic distribution ranges for the three nominal species inhabiting the AF (IUCN and Conservation International, 2015): *P. paulodutrai* (green; Pimenta and Juncá, 2004), *P. ramagii* (red; Juncá and Carnaval, 2004), and *P. vinhai* (orange; Peixoto and Pimenta, 2004). (B) Topology obtained by Bayesian inference based on mtDNA. Numbers at nodes represent posterior probability values. (C) Multiloci reticulated unrooted tree based on all combined nuclear loci. (For interpretation of the references to color in this figure caption, the reader is referred to the web version of this article.)

30 s, annealing temperature for 30 s and 72 °C for 45 s; plus a final extension step at 72 °C for 7 min. When necessary, both mitochondrial and nuclear PCR reactions were supplemented with 1 µL of BSA at 1 µg/µL. Annealing temperatures were as follows: 49 °C for *ND2*; 58 °C for *CRYBA*, *28S* and *MGC82783*. PCR purification was performed using polyethylene glycol 20% (PEG) precipitation (Sambrook et al., 1989) and sequenced in both DNA strands using the amplification primers at the Centro de Pesquisas Gonçalo Moniz - Fiocruz, Bahia, Brazil in an automated sequencer ABI PRISM 3500 (Applied Biosystems). We also gathered from the Genbank 100 sequences of ND2 available from Carnaval and Bates (2007). All the sequences generated in this study were deposited in GenBank under the following accession numbers:

ND2, MT166391-MT166564; 28S, MT160809-MT160867; CRYBA, MT166565-MT166629; MGC82783, MT166630-MT166673.

Electropherograms were inspected and assembled in contigs using CodonCode ALIGNER v. 6.0.2 (CodonCode Inc.). Sequences were aligned using MUSCLE (Edgar, 2004) in MEGA 5 (Tamura et al., 2011), and inspected and corrected visually, when necessary. Then, gametic phase of nuclear genes was determined using PHASE algorithm (Stephens, 2001) in DnaSP 5 (Librado and Rozas, 2009) under default settings, assuming 0.7 as a minimum probability. Individuals with probabilities below this threshold were removed from further analyses. We used the PHI test in SPLITSTREE4 (Huson and Bryant, 2006) to check for recombination in nuclear introns.

2.2. Phylogenetic analyses, species delimitation and genetic diversity

To investigate the mitochondrial structure of the clade, a phylogenetic reconstruction was performed for *ND2* including all samples using the Bayesian Inference (BI) with MrBayes 3.2.3 (Ronquist et al., 2012) at the CIPRES Science Gateway (Miller et al., 2010). We used GTR + I + G as the substitution model for *ND2*, as estimated by MrModeltest 2.2 (Nylander, 2004) based on the Akaike information criterion (AIC), in conjunction with PAUP* (Swofford, 1998). We performed two runs with 10 million generations and four Markov chain Monte Carlo (MCMC) each, with 10% of burn-in and parameters sampled every 1000th generations. Convergence of the runs was checked in Tracer v1.6 (http://beast.bio.ed.ac.uk/Tracer), making sure that all Effective Sample Size values were higher than 200. We used sequences of *Pristimantis cruentus* (Peters, 1873), *P. ridens* (Cope, 1866) and *P. fenestratus* (Steindachner, 1864) (Table S1) as outgroups. The monophyly of the Atlantic Forest *Pristimantis* clade was assumed based on the phylogeny from Canedo and Haddad (2012).

We implemented the Bayesian Phylogenetics and Phylogeography species delimitation method (BBP4) (Flouri et al., 2018) to infer putative species in the Atlantic Forest *Pristimantis* based in our multilocus dataset. Nuclear loci were used as phased alignments. As species we used the clades recovered by mtDNA BI tree (Fig. 1). The NCE clade was not included in this analysis because only mtDNA is available for this clade. We ran BPP with four prior configurations for population size (θ) and divergence time (τ) parameters proposed by Leaché and Fujita (2010): (1) large ancestral population sizes and ancient divergence [IG(3,0.2) for θ and τ]; (2) large population sizes and recent divergence [IG (3,0.2) for θ; and IG(3,0.002) for τ]; (3) small population sizes and recent divergences [IG(3,0.002) for theta and τ]; and (4) small populations and deep divergences [IG (3,0.002) for θ; and IG(3,0.2) fo τ]. We ran all combinations using algorithm 1 (automatic) available in BPP. We performed an A11 analysis (sensu Yang and Rannalla, 2014), which simultaneously delimits species and infers a species tree, using the species tree recovered in *BEAST as guide tree. Each run was performed with a burn-in phase of 10,000 iterations by sampling phase of 500,000 iterations, with frequency of two iterations per sampling.

To observe variation in each nuclear locus individually and the mitochondrial variation within each main clade, we obtained median joining haplotype networks using the software PopART (Leigh and Bryant, 2015). We used POFAD V. 1.03 (Joly and Bruneau, 2006) to generate a combined multilocus nuclear distances matrix, which was based on the standardized distances and the genpofad index. The resulting matrix was used to generate a NeighborNet reticulated unrooted tree in SPLITSTREE V. 4.6 (Huson and Bryant, 2006). As we were not able to obtain sequences of nuclear loci for samples from northern state of Ceará, this group was not included in these analyses. Haplotype (HD) and nucleotide (π) diversities were obtained in DnaSP 5.1. Pairwise genetic distances (p-distance) among main lineages were estimated for ND2 on MEGA 5 (Tamura et al., 2011).

2.3. Species tree and divergence time

To estimate divergence times between lineages, we combined information of all gene trees in a single tree by applying the Bayesian multispecies coalescent method of species tree in *BEAST (Bayesian Inference of Species Trees from Multilocus Data; Heled and Drummond, 2010) using BEAST 1.8.4 (Drummond et al., 2012) the CIPRES Science Gateway. We also obtained divergence times using the multispecies coalescent method of *BEAST. We generated the input file in BEAUTi considering all four markers. The mtDNA lineages (Fig. 1) were considered as putative species in the analysis. The NCE lineage was not included because only mtDNA was generated. The best fit model for each locus was estimated by MrModeltest: *ND2* – GTR + I + G; *28S* – HKY + I; *CRYBA* and *MGC82783* – HKY + G. Piecewise linear and constant root were used as population size model priors. As there is no fossil available for the group we used the substitution rate for *ND2* of 0.00957 mutation/site/million year (Crawford, 2003) to calibrate the tree. Then, we allowed BEAST estimates rates for nuclear loci under a default lognormal distribution. We performed two independent runs with 200 million generations of MCMC, sampling every 10,000 generations and using a 10% of burn-in. We checked for convergence between runs using Tracer v1.6, making sure that ESS values were higher than 200 for all parameters. The resulting trees were combined using TreeAnotator. We visualized the consensus tree and posterior sampled trees in DensiTree 2.2.1 (Bouckaert, 2014).

3. Results

3.1. Phylogenetic analyses, species delimitation and genetic diversity

We obtained a dataset with 1797 bp: 688 bp of *ND2* (n = 273, 361 polymorphic sites), 197 bp of *CRYBA* (n = 65, 40 polymorphic sites), 732 bp of *28S* (n = 59, 28 polymorphic sites) and 180 bp of *MGC82783* (n = 45, 57 polymorphic sites) (Table S1). The PHI test did not find statistically significant evidence for recombination in all nuclear loci (*CRYBA*, $p = 0.589$; 28S, $p = 0.073$; in *MGC82783* there were too few informative characters to implement the test). After allelic reconstruction by PHASE, haplotypes with low probability ($p < 0.7$) were removed from further analyses (6 samples of *28S* and 10 samples of *MGC82783*).

The phylogenetic inference based on mtDNA data revealed the Atlantic Forest *Pristimantis* encompassed nine well supported main clades (Fig. S1): (01) Northern Ceará clade (hereafter, NCE clade) consisting of the samples from enclaves of Atlantic Forest on Ceará state; (02) Jacobina clade (JCB), with samples from an Atlantic Forest enclave at Serra de Jacobina, Bahia state; (03) Rio de Contas clade (RDC), from two municipalities at Chapada Diamantina, Bahia state; (04) *vinhai* clade (VNH), composed of samples from southern Bahia state, including the type locality of *P. vinhai* (municipality of Ilhéus); (05) *paulodutrai* clade (PDT), composed of samples from Bahia and Sergipe states, including the type locality of *P. paulodutrai* (municipality of Ilhéus); (06) São Francisco River clade (SFR), with samples located north of the São Francisco River; (07) *ramagii* clade (RMG), with samples from Pernambuco, Paraíba and Rio Grande do Norte states, including the type locality of *P. ramagii* (municipality of Igarassu, Pernambuco state); (08) pseudoramagii clade (PRA), with samples from Bahia, Alagoas and Pernambuco states; and (09) Southern Bahia clade (SBA), from southern Bahia state (Fig. 1).

Nuclear networks partially recovered the mitochondrial structure. The network for *CRYBA* identified 34 haplotypes (Fig. 2C), in which only JCB and RDC clades did not share haplotypes with other clades. Haplotype network for *28S* identified 16 haplotypes (Fig. 2B), in which only RMG and JCB clades did not share haplotypes with other clades. *MGC82783* network identified 31 haplotypes (Fig. 2A), in which only two haplotypes were shared between clades (SFR/RMG and PDT/SBA). The combined nuclear multilocus network based on POFAD matrix

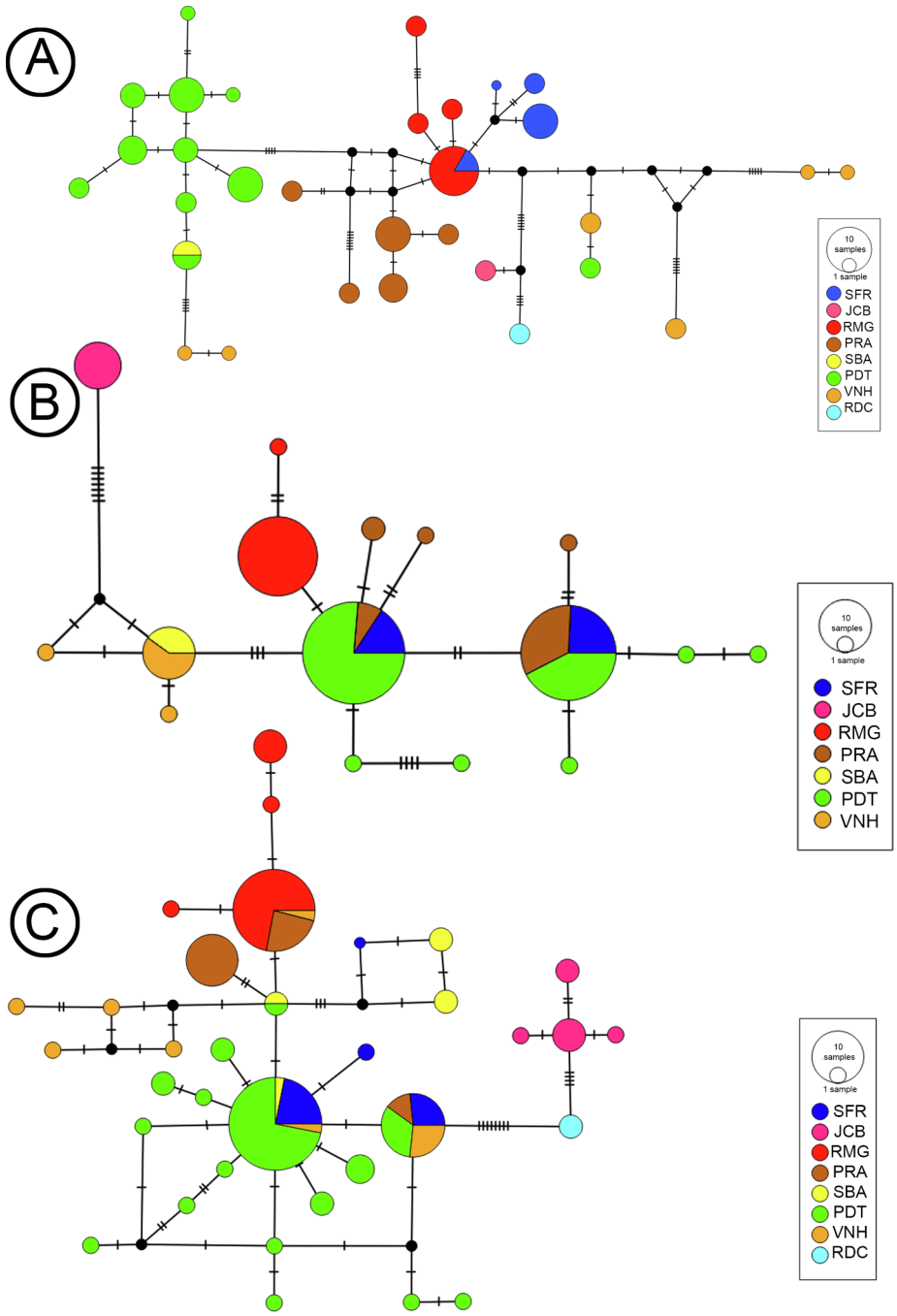

Fig. 2. Haplotype networks for nuclear markers. Each circle represents a different haplotype. Circle sizes are proportional to haplotype frequencies. The number of mutational steps between haplotypes is represented at the line connecting the haplotypes. Colors refers to lineages and are in accordance with Fig. 1. (A) Haplotype network for the nuclear intron *X. laevis* MGC82783 protein Intron2 (MGC82783). (B) Haplotype network for the nuclear intron 28S ribosomal RNA (28S). (C) Haplotype network for the nuclear intron Beta-crystallin (CRYBA).

recovered three main clusters (Fig. 1). The first cluster grouped samples mostly from clades PDT, SFR, RMG and PRA, but also with samples from VNH and SBA lineages. The second cluster is composed of samples from JCB and RDC, with the largest divergence in the network. The third cluster comprises samples mostly from clade VNH, but also with samples from PDT and SBA lineages. We also observed high levels of variation within most main lineages of the Atlantic Forest *Pristimantis* (Fig. S2, at supplementary material). Only RDC and JCB did not show high variation, but this is most likely a consequence of the low number of samples for these clades.

BPP results with all of the four prior combinations indicated presence of distinct species in the Atlantic Forest *Pristimantis*. Notwithstanding, runs with distinct prior parameters delimited distinct numbers of species. RDC and JBC lineages were consistently recovered as species in all runs. However, the runs using the prior combination 1 (large ancestral population sizes and ancient divergence) and prior combination 3 (small populations sizes and recent divergences) clustered lineages PRA, RMG, SFR and PDT into a single species. The run using the prior combination 2 (large population sizes and recent divergence) recovered the eight lineages into a single species with probability 1.0, which is accordance with results of species tree and mtDNA phylogeny. Lastly, the run using the prior 4 combination (small populations and deep divergences) grouped the lineages VNH and SBA into a single species (see Fig. 3). All runs exhibited high values of

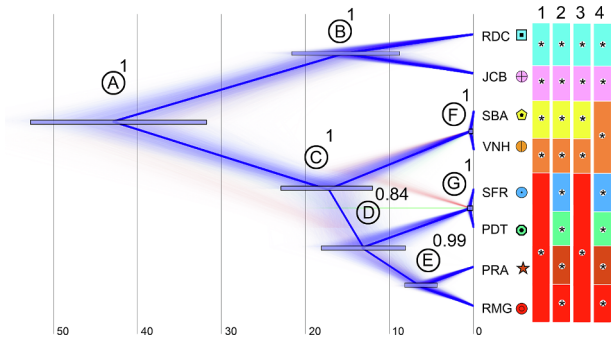

Fig. 3. Multispecies coalescent species tree chronogram from both mitochondrial and nuclear dataset. Numbers at nodes represent posterior probabilities. Faded lines represent each tree sampled, the consensus tree is represented by the thick blue line. Circled letters at nodes represent divergence times and 95% of high posterior density of divergence times between clades at Table 2. Colored bars indicate species delimited by BPP. Numbers above bars indicate run with distinct prior set. Asterisks indicates species identified with probability of 1.0. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

probability (1.0)

Haplotype diversities for *ND2* ranged from 0.28 in RDC to 0.97 in SBA clade. The highest nucleotide diversity was found in VNH and the lowest in JCB clade (Table 1). Pairwise genetic distances between clades ranged from 0.085 (RMG and PRA) to 0.325 (NCE and SFR) (Supplementary data 2, Supplementary material).

3.2. Species tree and divergence times

The species tree generated by *BEAST showed similar relationships as the ones recovered on mtDNA phylogeny (except at basal nodes in which RDC and JCB form a clade in species tree), with strong support for most nodes (Fig. 3, Table 2). The NCE clade was not included on this analysis due to the lack of nuclear sequences. Our estimation of divergence time generated by *BEAST indicated that initial split in the Atlantic Forest *Pristimantis*, which separates RDC and JCB from remaining clades, took place in the Eocene (c.a. 42.93 mya; Fig. 3, Table 2). The split between RDC and JCB occurred in the middle Miocene, at c.a. 15.56 mya. The cladogenesis among other clades occurred at early Miocene, at *c.a.* 17.11 mya. The RMG and PRA lineages diverged during the middle Miocene, c.a. 13.2 mya. The later divergence of the remaining clades (VNH and SBA: 0.28 mya; PDT and SFR: 0.35 mya) took place at the late Pleistocene.

4. Discussion

Our comprehensive geographic sampling and multilocus dataset allowed us to identify nine distinct evolutionary lineages, more than expected based on the previously recognized species for the Atlantic Forest. Intriguingly, divergence time estimates revealed an ancient diversification that took place during the Eocene and Miocene, with few splits in the late Pleistocene.

Table 1
Summary statistics for the mitochondrial clades and for nuclear markers.

Fragment	Group	H	Mean Hd (95% CI)	π	N
ND2	All	154	0.99 (0.99–0.99)	18.5	273
	NCE	15	0.95 (0.92–0.97)	1.1	26
	JCB	2	0.40 (0.17–0.63)	0.05	5
	RDC	2	0.28 (0.09–0.47)	0.5	7
	VNH	38	0.98 (0.97–0.99)	8.3	59
	PDT	26	0.95 (0.93–0.97)	5.4	42
	SFR	8	0.82 (0.77–0.88)	4.6	20
	RMG	40	0.97 (0.96–0.97	2.7	83
	PRA	12	0.94 (0.9–0.97)	4.4	18
	SBA	11	0.97 (0.93–1)	4.2	13
28S	All	15	0.79 (0.77–0.80)	0.4	120
CRYBA	All	34	0.88 (0.86–0.89)	2.3	126
MGC82783	All	30	0.95 (0.94–0.96)	4.1	88

Table 2
Divergence times (mean) and confidence intervals [95% of high posterior density (HPD)] for the nodes labeled on Fig. 3.

Node	Time (Mya)	95% of HPD
A	42.93	32.12–52.86
B	15.56	8.74–21.63
C	17.11	12.02–22.99
D	13.2	8.09–18.02
E	6.24	4.30–8.14
F	0.28	0.12–0.49
G	0.35	0.12–0.65

4.1. Systematics and species limits in the Atlantic Forest Terrarana

The diversity of *Pristimantis* is concentrated on northwestern South America (Hedges et al., 2008), currently with only three eastern nominal species, all endemic to the Atlantic Forest (Canedo and Haddad, 2012). Our data revealed that the Atlantic Forest *Pristimantis* encompasses nine well supported divergent lineages, possibly representing three times more species than previously recognized for the genus in the region. Given the old time since the split of Atlantic Forest and Western species of Terraranae (Canedo and Haddad, 2012) and the remarkable diversity within the group, the relatively few species of Atlantic Forest *Pristimantis* currently described was somewhat puzzling. Yet, direct developing frogs have climatically restricted distributions (Gomez-Mestre, et al., 2012; Lion et al., 2019) which, along with egg and clutch sizes, impose severe limits to the persistence of these species (Liedtke et al., 2017). Therefore, past and current climates may have been responsible for the current distribution and relatively small number of *Pristimantis* species in the Atlantic Forest. Nevertheless, our results show a great hidden diversity.

Only clades NCE and RDC did not occur in sympatry with another clade. Clades RMG and SFR only occurred in sympatry with each other (municipality of Mamanguape, state of Paraíba). PRA clade occurred in sympatry with clades JCB (municipality of Miguel Calmon), PDT (municipality of Catu) and SBA (municipality of Elísio Medrado), all in the state of Bahia. VNH clade occurred in sympatry with clades SBA (municipality of Ilhéus, Guaratinga and Elísio Medrado) and PDT (municipalities of Ilhéus, Una and Uruçuca), all in the state of Bahia. PDT clade occurred in sympatry with clades PRA (municipality of Catu and Jequié), VNH (municipalities of Ilhéus, Una and Uruçuca) and SBA (municipality of Ilhéus), all in the state of Bahia.

The clade VNH is distributed throughout the range of *P. vinhai*, including its type locality, and is restricted to Bahia state. The few records of *P. vinhai* above the São Francisco River (Morato et al., 2011; Santana et al., 2016; dos Santos et al., 2016) likely represent misidentifications, given the complex morphology of the species group. We also found a sister clade to VNH, the SBA clade, that is not currently described as a nominal species. The VNH and SBA clades may occur in sympatry but based on our field observations individuals from these clades differ in activity time, SBA vocalizes only at twilight while VNH is more active after sunset. The genetic distance found between SBA and VNH was similar to what was found between VNH and the other lineages associated to the three valid species (PDT and RMG) (Table S2). They are likely two distinct species. Furthermore, we found *P. paulodutrai* to be monophyletic (clade PDT), but samples from Salvador (Northern group) and Ilhéus (Southern group) were placed on different haplogroups, corroborating previous karyotypic studies (Siqueira et al., 2008; Napoli et al., 2009). The VNH and PDT clades co-occur in three localities

(municipalities of Igrapiúna, Ilhéus and Uruçuca, state of Bahia). We expected to find at least two lineages around the municipality of Ilhéus, since it is the type locality for two of the nominal species of *Pristimantis* of the Atlantic Forest (*P. paulodutrai* and *P. vinhai*), but these species are morphologically and acoustically distinguishable (Fonseca, 2011). Clades VNH and PDT occupied Bahia state in different times, and they correspond to morphologically different species. The two lineages are sympatric but there is no haplotype sharing on three of the four markers used, a strong evidence for these clades to comprise different species, what is reinforced by the species delimitation analysis and the genetic distance between both clades (0.21).

We could also expect more than one lineage above the São Francisco River, since Carnaval and Bates (2007) reported differences in call structure and microhabitat use between two distinct haplotype groups occurring sympatrically in the municipality of Mamanguape, state of Paraíba. Here, with a more comprehensive sampling, sequences from that were distributed among three clades: NCE, RMG and SFR, with RMG and SFR occurring sympatrically in Mamanguape. Our field observations endorse that RMG and SFR differ acoustically, and also behaviorally, as we registered calling males of SFR clade vocalizing from herbaceous vegetation or on tree leaves at about 200–400 cm high from the ground at the edge of ombrophilous forest on montane environment (531 m above sea level), in contrast to individuals from the RGM clade, which calls on the ground with leaf litter, perched on herbaceous vegetation in shrub areas or in spaced tree areas at about 5–60 cm high from the ground at the edge of open ombrophilous forests (16–530 m a.s.l.).

The species delimitation inference by BPP indicated between 5 and 8 putative species in the Atlantic Forest *Pristimantis*. However, the run assuming large population sizes and recent divergence recovered the same number of species observed in *BEAST and mtDNA tree. The possible presence of gene flow and/or old and recent divergence across diversification (Miocene to Pleistocene splits) might have influenced the species delimitation of BPP. Further studies of species status and delimitation should address these issues. The NCE Clade was not included in this analysis due to the absence of nuclear data, but considering its high genetic distance from other clades (32.5 to 25.2% of divergence), we might assume this lineage as a possible new species in this group.

Due to the co-occurrence of divergent genetic lineages and high morphologic and acoustic variation in the Atlantic Forest *Pristimantis*, we believe further studies are needed to correctly delimit species of *Pristimantis* in the northern Atlantic Forest. Other direct development frogs have been shown to harbor ancient and highly divergent lineages under few recognized taxonomic names, such as in the southern Atlantic Forest *Ischnocnema* (Gehara et al., 2013). Other lines of evidence, such as advertisement calls, internal morphology, and natural history, under an integrative framework, is needed to elucidate the taxonomy of the *Pristimantis* from Atlantic Forest.

4.2. Diversification history of Pristimantis in the northern Atlantic Forest

The *Pristimantis* clade in South America split from other Terraranae from 26 to 50 mya, with a rapid diversification starting at 24 mya, likely associated to the Andean uplift (Heinicke et al., 2007). Although the most significant uplift of the Andes occurred in the late Miocene, the process began during the Paleogene, with also great mountain building on late Oligocene (Hoorn, 2010). The earlier split in the *Pristimantis* from the Atlantic Forest (32.13–52.86 mya, Fig. 3) is in accordance with the ancient diversification recovered by Heinicke et al. (2007). Indeed, other studies have shown ancient lineages and clades endemic to the Atlantic Forest for birds (Derryberry et al., 2011; Fjeldså et al., 2012) and frogs (Fouquet et al., 2012). Our results evidenced that the northern region of the Atlantic Forest also holds species and lineages that resemble the early bursts of the diversification history, especially in forest enclaves within the Caatinga biome.

The oldest split in the Atlantic Forest *Pristimantis* comprehend the isolation and divergence of lineages from forest enclaves, within the current distribution of the Caatinga seasonally dry woodlands, and lineages from the coastal Atlantic Forest, at c.a. 42.9 mya (Fig. 3). The forest enclaves studied herein comprehend scattered patches at higher elevations, the Chapada Diamantina in Bahia state (RDC and JCB) and forest enclaves in Ceará state (NCE). The other lineages of *Pristimantis* (SFR, PDT, SBA, PRA, RMG and VNH) comprise a clade restricted to the coastal Atlantic Forest, except for the occurrence of PRA in both Chapada Diamantina and coastal Atlantic Forest. Nevertheless, enclave lineages were recovered as monophyletic only in the species tree (Fig. 3), which did not include NCE. Despite this, the oldest split that isolated coastal and enclaves' lineages can be associated to the climatic oscillations during the Eocene-Oligocene. During early Eocene the earth experienced pronounced warming, as in the greenhouse period that largely increase levels of CO^2 at c.a. 56 mya (Kiehl et al., 2018), followed by a period of cooling during transition of Eocene-Oligocene (c.a. 34 mya) that drove ice sheets grow over most of Antarctica and central Asia aridification (Bowen, 2007). Yet, divergence between lineages within enclaves clade and in early nodes in coastal clade during the Miocene might also be associated with climatic oscillations during this period, with a temperature reduction at the middle Miocene (Zachos et al., 2001). These events might have contributed to the establishment and the species build up in the early history of the Atlantic Forest. However, as gene flow can overestimate and underestimate population sizes and divergence times, respectively, of species tree inference (Leaché et al., 2014), these results might be assumed with cautionary due to possible gene flow among lineages.

The later splits between lineages VNH and SBA and between PDT and SFR during the late Pleistocene seem to have been affected by Quaternary climate changes that isolated and connected population during cyclical periods of forest retraction (at cooler periods) and expansion (at warmer periods) (Carnaval et al., 2009, 2014; Silveira et al., 2019). Additionally, it is worth noting that PRA lineage also is distributed in Chapada Diamantina, but the lower divergence of coastal population indicates that this group may have recently dispersed to this region, driven by late Pleistocene climatic dispersal routes between coastal and Chapada Diamantina forests (Silveira et al., 2019). Further studies combining genetic, ecological and modelling tools on these earlier diverged groups might shed light on this issue.

CRediT authorship contribution statement

Camila Costa Trevisan: Conceptualization, Formal analysis, Investigation, Writing - original draft, Writing - review & editing, Visualization. **Henrique Batalha-Filho:** Conceptualization, Formal analysis, Writing - original draft, Writing - review & editing, Supervision, Resources, Funding acquisition. **Adrian Antonio Garda:** Resources, Writing - original draft, Writing - review & editing. **Lucas Menezes:** Formal analysis, Investigation, Writing - original draft, Writing - review & editing. **Iuri Ribeiro Dias:** Resources, Writing - original draft, Writing - review & editing. **Mirco Solé:** Resources. **Clarissa Canedo:** Conceptualization, Writing - original draft, Writing - review & editing. **Flora Acuña Juncá:** Resources. **Marcelo Felgueiras Napoli:** Conceptualization, Formal analysis, Writing - original draft, Writing - review & editing, Supervision, Resources, Funding acquisition, Project administration, Visualization.

Acknowledgements

We are grateful to Célio F. B. Haddad (CFBH), Gustavo H. C. Vieira (CHUFPB), Robson Ávila (URCA), Tamí Mott (MUFAL), José P. Pombal Jr (MNRJ) and Ednilza Maranhão (UFRPE) for providing tissue samples for the molecular analyses; Rafael O. Abreu (UFBA) and Jamile Santos-Nascimento (UFBA) for the excellent field assistance; the Plataforma de Sequenciamento at Centro de Pesquisa Gonçalo Moniz (Fiocruz) for

sequencing the samples. We thank the Instituto Brasileiro do Meio Ambiente e dos Recursos Naturais Renováveis (IBAMA), Instituto Chico Mendes de Conservação da Biodiversidade (ICMBio, license 12988-1) and the local administration of the Parque Estadual das Sete Passagens - PESP (José M. Pereira – "Zélis") for permissions to collect anurans. This work was supported by the Fundação de Amparo à Pesquisa do Estado da Bahia (Fapesb) and the Secretaria do Meio Ambiente (SEMA), under the research projects "Padrões e processos da diversidade de anfíbios na Bahia: influência das mudanças climáticas e propostas de conservação" (TO PAM0005/2014) and FAPESB JCB0026/2016, developed by the "Rede Baiana de Pesquisa sobre Anfíbios – RBPA"; RED0045/2014; and the Conselho Nacional de Desenvolvimento Científico e Tecnológico - CNPq (Proc. No. 458647/2014-4, 443249/2014). MFN, HBF, AAG and FAJ thank CNPq for productivity grants (305849/2015-8 and 310490/2018-9, 307037/2018-5, 310942/2018-7, 305704/2013-3, respectively). LMS and CCT thanks the Fapesb for doctorate's fellowships (grants BOL0494/2016, BOL3531/2014 and BOL2582/2016, respectively). IRD was supported by CNPq (Process: 167387/2017-0 and 155198/2018-1) and Boticário Group Foundation for Nature Protection, under the research project "Diversity and genetic structure of the amphibians of the Central Corridor of the Atlantic Forest of southern Bahia" (Project No. 0991_20132).

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.ympev.2020.106811.

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